

Project Development Phase

Date	13 July 2026
Team ID	
Project Name	Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau
Maximum Marks	4 Marks

Data Preprocessing Steps

Step 1: Import the Dataset

- Imported the Housing Sales dataset (~21,609 records) into Tableau Public.
- Verified that all records and fields (Sale Price, Bedrooms, Bathrooms, Living Area, Lot Area, Floors, Waterfront, View, Grade, Condition, Year Built, Year Renovated, Zipcode, etc.) were imported successfully.
- Confirmed that the dataset was ready for preprocessing.

Step 2: Check for Missing Values

- Examined the dataset for null or blank values.
- Verified that important fields such as Sale Price, Bedrooms, Bathrooms, Living Area and Zipcode contained valid information.
- Missing values were handled appropriately to ensure accurate analysis.

Step 3: Remove Duplicate Records

- Checked the dataset for duplicate property records using the property ID.
- Removed duplicate entries to prevent inflated counts and incorrect market insights.
- Ensured that every property sale was represented only once.

Step 4: Validate Data Types

Ensured that each attribute had the correct data type.

- Property ID → String
- Sale Price → Number
- Bedrooms → Number (Integer)
- Bathrooms → Number (Decimal)
- Living Area (Sqft) → Number
- Lot Area (Sqft) → Number
- Floors → Number (Decimal)
- Waterfront → Categorical
- View → Categorical
- Grade / Condition → Categorical
- Year Built → Number
- Year Renovated → Number
- Zipcode → Geographic/String

Step 5: Verify Data Consistency

- Checked for invalid or unrealistic values such as zero/negative sale price, zero bedrooms or bathrooms, or unrealistic living-area values.
- Standardized zipcode groupings, renovation-status categories, and grade/condition labels.
- Ensured consistency across all categorical fields before visualization.

Step 6: Prepare Calculated Fields

Created calculated fields required for dashboard analysis, including:

- Total Properties
- Average Sale Price
- Total Living / Basement Area
- House Age (current year – year built)
- Years Since Renovation
- Renovation Status (Renovated / Not Renovated flag)
- Sale Price Bins
- Zipcode Group
- KPI Measures for Dashboard

Step 7: Final Clean Dataset

- Verified that the cleaned dataset was complete and suitable for analysis.
- Confirmed that all preprocessing activities were completed successfully.
- The dataset was finalized for dashboard creation and business intelligence visualizations in Tableau Public.

Business Questions with Visualization

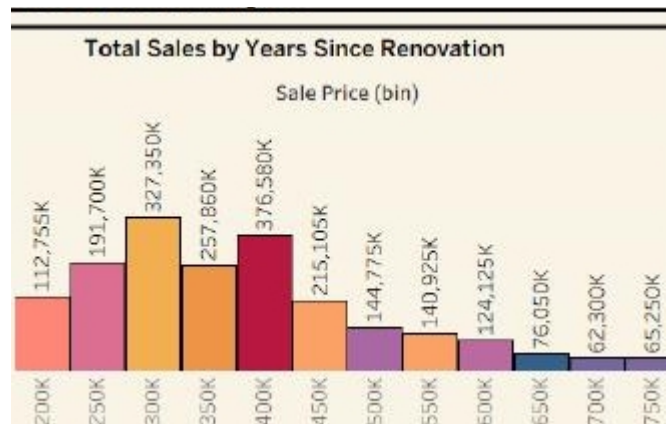
S.No.	Business Question	Visualization
1	How many housing sale records are available for analysis after data preprocessing, what is the average sale price, and what is the total basement area across properties?	KPI Summary Panel (KPI Cards)
2	How do total sales vary by the number of years since renovation?	Total Sales by Years Since Renovation (Bar Chart)
3	How is house age distributed across renovated vs. non-renovated properties?	Distribution of House Age by Renovation Status (Pie Chart)
4	How does house age vary by number of bathrooms, bedrooms and floors?	House Age Distribution by Bathrooms, Bedrooms & Floors (Bar Chart)

3. Visualizations:

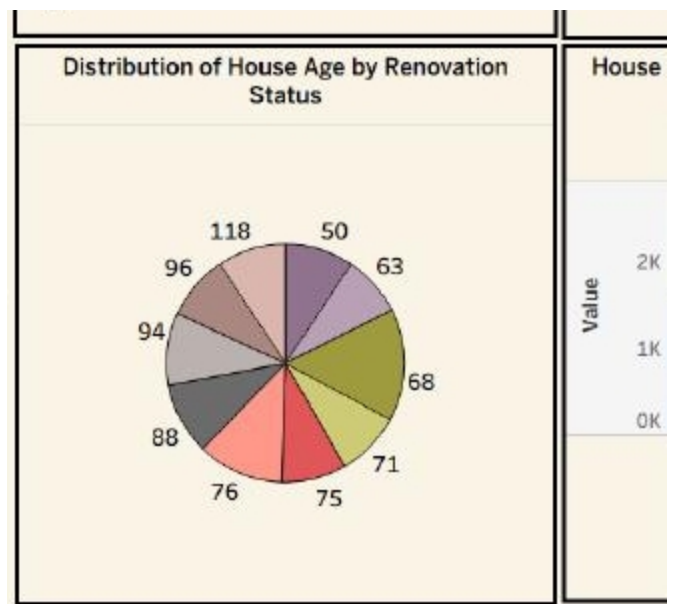
KPI Summary Panel



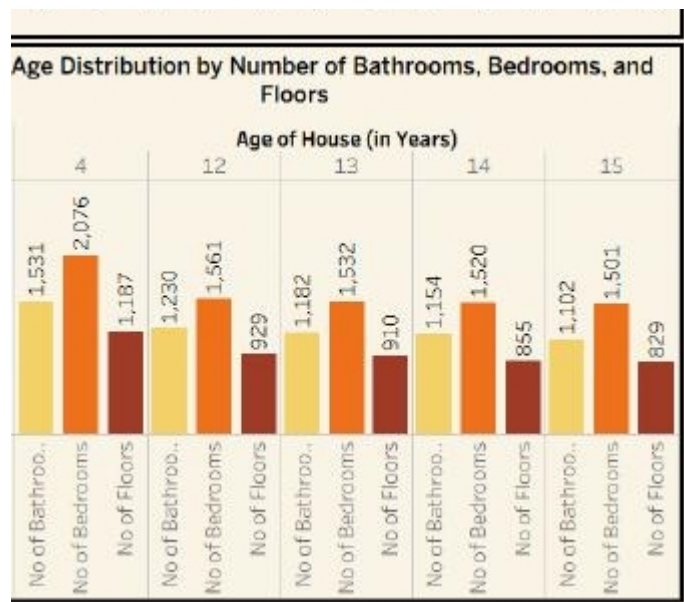
Total Sales by Years Since Renovation



Distribution of House Age by Renovation Status



House Age Distribution by Bathrooms, Bedrooms & Floors



Dashboard:

Comprehensive House Data Analysis — published Tableau Public dashboard combining the KPI summary panel with all supporting visualizations.

